



Foundation Certificate in Higher Education

Module : DOC 333 – Computer programming

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Topic : Transportation system of ‘Kingdom of Miranda’

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1. Problem Specification

1.1. Introduction:

The "DropMe™ Cab Service Console Application" is designed to facilitate passengers in the Kingdom of Miranda to view trip details, calculate the fare, and generate invoices for their cab rides. The application operates via a command-line interface and allows passengers to choose different transport modes and apply promo codes (if available) to receive discounts on their trips.

1.2.Features:

The DropMe™ Cab Service Console Application offers the following features:

- Calculate and display the fare for traveling between different cities.
- Allow passengers to choose between Trishaws, Cars, or Vans as their preferred transport mode.
- Provide the ability to apply promo codes to receive discounts on the total fare.
- Generate invoices in text files for each trip, including trip details, applied promo codes, and the final payment amount.
- Optionally apply random fixed discounts for some passengers.

1.3.Command Types:

The following command types can be issued to interact with the application:

Command	Description
C:\Users\Nishan>dm alvin razi	Shows the price between the two cities and generates an invoice file for the trip
C:\Users\Nishan>dm alvin razi /pro2	Shows the price between the two cities after applying the promo code. The promo code amount is deducted from the final bill value. This will generate an invoice file for the trip

	with promo details. 2 KMD is reduced here.
C:\Users\Nishan>dm /price	Show the full price plan for the whole country (for all 3 vchiles)
C:\Users\Nishan>dm alvin razi /c	Shows the price between the two cities and generates an invoice file for the trip. This uses a car (/c for car while /v for a van. Default is Trishaw)
C:\Users\Nishan>dm alvin razi /pro10 /v	Shows the price between the two cities while apply a 10 KMD reduction to total bill. The rider prefers a van.
C:\Users\Nishan>dm /?	Show how this “dm” command functions

Table 1 - Command Types

1.4.Promo Codes:

The application supports the following promo codes, where the number represents the KMDs reduced from the total bill when applied:

pro1, pro2, pro3, pro4, pro5, pro6, pro7, pro8, pro9, pro10, pro11, pro12, pro13, pro14, pro15

1.5.Invoice Generation:

Every time a passenger uses the application, an invoice is saved in a text file in the following format: <date> <time>_<random_number>.txt (e.g., "2023-06-26 06_30_29_4526.txt"). The invoice file contains the following details:

- Date and Time of the trip.
- Start and End Locations.
- Amount (Original fare before applying any promo or discount).
- Promo (Amount reduced due to the applied promo code, if any).
- Random Reduction (Amount reduced due to the random fixed discount, if any).
- Final Payment (The total amount the passenger needs to pay after all reductions).

1.6.Operating Guidelines:

- Passengers can enter city names in any letter case (upper or lower) since the application is case-insensitive.
- If a promo code is provided, it will be checked for validity, and the corresponding discount will be applied if it exists in the list of promo codes.
- Passengers can choose the preferred transport mode by providing /c for Car or /v for Van. The default transport mode is Trishaw if none is specified.
- Passengers can receive random fixed discounts on the fare. This feature is independent of promo codes and applied randomly by the application.

1.7.Conclusion:

The "DropMe™ Cab Service Console Application" provides a user-friendly interface for passengers in the Kingdom of Miranda to calculate trip fares, apply discounts via promo codes, and generate invoices for their cab rides. With its easy-to-use commands, passengers can view pricing, choose transport modes, and enjoy occasional random discounts during their trips. The application streamlines the cab booking process, making it efficient and convenient for passengers to plan their travels within the country.

2. Full python code.

```
import sys
from datetime import datetime
import random
import math

# Cities price list for traveling
cities_price_list = {
    'Alvin': {'Alvin': 0, 'Jamz': 20, 'Razi': 40, 'Mali': 40, 'Zuhar': 20},
    'Jamz': {'Alvin': 20, 'Jamz': 0, 'Razi': 20, 'Mali': 40, 'Zuhar': 40},
    'Razi': {'Alvin': 40, 'Jamz': 20, 'Razi': 0, 'Mali': 20, 'Zuhar': 40},
    'Mali': {'Alvin': 40, 'Jamz': 40, 'Razi': 20, 'Mali': 0, 'Zuhar': 20},
    'Zuhar': {'Alvin': 20, 'Jamz': 40, 'Razi': 40, 'Mali': 20, 'Zuhar': 0}
}

# Transport modes
vehicle_prices = {
    'trishaw': 1,
    'car': 2,
    'van': 3
}

# Promotional codes
promo_codes = {
    'pro1': 1,
    'pro2': 2,
    'pro3': 3,
    'pro4': 4,
    'pro5': 5,
```

```
'pro6': 6,  
'pro7': 7,  
'pro8': 8,  
'pro9': 9,  
'pro10': 10,  
'pro11': 11,  
'pro12': 12,  
'pro13': 13,  
'pro14': 14,  
'pro15': 15  
}
```

```
# Defined transport service
```

```
def transport_service():
```

```
    while True:
```

```
        city_name = ['Alvin', 'Jamz', 'Razi', 'Mali', 'Zuhar']
```

```
        transport_mode = ['Trishaw', 'Car', 'Van']
```

```
        # Welcome message
```

```
        print("-----Welcome to DropMe Cab service-----")
```

```
        print("The available cities :", city_name)
```

```
        print("Available transport modes :", transport_mode)
```

```
        # Get trip details from the user
```

```
        current_city, destination_city, transport_method = get_trip_details()
```

```
        # Calculate trip price
```

```
        prices = calculate_prices(transport_method, current_city, destination_city)
```

```
        print("The gross price for your trip is:", prices, "KMD")
```



```

# Apply promos and calculate final payment
promo_type, total = operate_promos(prices)
print("Total payment :", total, "KMD")

# Save trip details to a text file
save_details(current_city, destination_city, prices, promo_type, total)

# Ask user if they want to use the service again
user_choice = input("Do you want to use the service again? (yes/no): ").lower()
if user_choice != 'yes':
    break

# Define promotion codes
def operate_promos(total):
    # Ask if the user has a promotional code
    promo_status = input("Do you have a promo code? (yes/no): ").lower()

    if promo_status == "yes":
        promo = input("Enter promo code here: ")
        while promo not in promo_codes:
            promo = input("Invalid promo code. Enter a valid promo code: ")

    # Apply promo code and calculate total payment
    discount = promo_codes[promo]
    total = max(total - discount, 0)
    return "promo", total
else:
    # Randomly apply a 5 KMD discount with a 33.33% chance
    if random.randint(1, 3) == 1:
        total = max(total - 5, 0)

```

```

        return "random", total
    return "none", total

# Defined function to get the trip details
def get_trip_details():
    current_city = input("Enter the Start Location: ")
    destination_city = input("Enter the End Location: ")
    transport_method = input("Enter the transport mode (Trishaw/Car/Van): ").lower()

    # Ensure valid transport method is selected
    while transport_method not in vehicle_prices:
        transport_method = input("Invalid transport method. Enter a valid method (Trishaw/Car/Van): ").lower()

    return current_city, destination_city, transport_method

# Defined function to calculate prices based on the selected transport method
def calculate_prices(transport_method, current_city, destination_city):
    price = cities_price_list[current_city][destination_city]
    return price * vehicle_prices[transport_method]

# Defined function to save trip details to a text file
def save_details(departure, destination, prices, promo_type, total):
    now = datetime.now()
    date_time = now.strftime("%Y-%m-%d %H_%M_%S")
    random_number = random.randint(1000, 9999)
    file_name = f"{date_time}_{random_number}.txt"

    with open(file_name, "w") as file:
        file.write(f"Date: {now.date()}\n")

```

```
file.write(f"Time: {now.time()}\n")
file.write(f"Start: {departure}\n")
file.write(f"End: {destination}\n")
file.write(f"Amount: {prices} KMD\n\n")
file.write(f"Promo: {prices - total if promo_type == 'promo' else 0} KMD\n")
file.write(f"Random Reduction: {prices - total if promo_type == 'random' else 0} KMD\n")
file.write(f"Final payment: {total} KMD\n")

print("\n-----Receipt successfully printed-----")
print("Receipt name:", file_name)
print("\n")

# Run the transport service
if __name__ == "__main__":
    transport_service()
```

3. Algorithm

1. Import required modules:

Import the sys module to handle system-specific functions (not used in the provided code).

Import the datetime module to work with date and time.

Import the random module to generate random numbers.

Import the math module (not used in the provided code).

Define a dictionary cities_price_list to store the prices for traveling between different cities. The keys represent the starting city, and the nested dictionaries contain destination cities as keys with their corresponding prices.

Define a dictionary vehicle_prices to store the prices for different transport modes. The keys represent the transport mode (e.g., Trishaw, Car, Van), and the values represent the price multipliers for each mode.

Define a dictionary promo_codes to store promotional codes and their corresponding discounts.

Define a function transport_service():

Display a welcome message, listing available cities and transport modes.

Get trip details from the user (current city, destination city, and transport method).

Calculate the trip price based on the selected transport method and the cities' prices.

Apply promotions (if any) and calculate the final payment.

Save trip details to a text file.

Ask the user if they want to use the service again. If not, break the loop and exit the function.

Define a function `operate_promos(total)` to handle promotional codes:

Ask the user if they have a promo code.

If yes:

Ask the user to enter the promo code until they enter a valid one from the `promo_codes` dictionary.

Apply the promo code's discount to the total price and return the promo type and the new total.

If no:

Randomly decide to apply a fixed 5 KMD discount to the total price.

Return the promo type (promo, random, or none) and the new total.

Define a function `get_trip_details()` to get the trip details from the user:

Ask the user to enter the starting location (current city), ending location (destination city), and transport mode.

Ensure a valid transport mode is selected by repeatedly asking until the user enters a valid one.

Return the current city, destination city, and transport method.

Define a function `calculate_prices(transport_method, current_city, destination_city)` to calculate the trip price:

Use the provided dictionaries `cities_price_list` and `vehicle_prices` to look up the prices for the selected transport mode and the trip between the current city and the destination city.

Calculate the final price as the product of the base price and the transport mode price multiplier.

Return the final price.

Define a function `save_details(departure, destination, prices, promo_type, total)` to save trip details to a text file:

Generate a filename based on the current date, time, and a random number to avoid overwriting previous files.

Open the file in write mode and write the trip details and payment information.

Print the receipt's filename, indicating that the receipt was successfully printed.

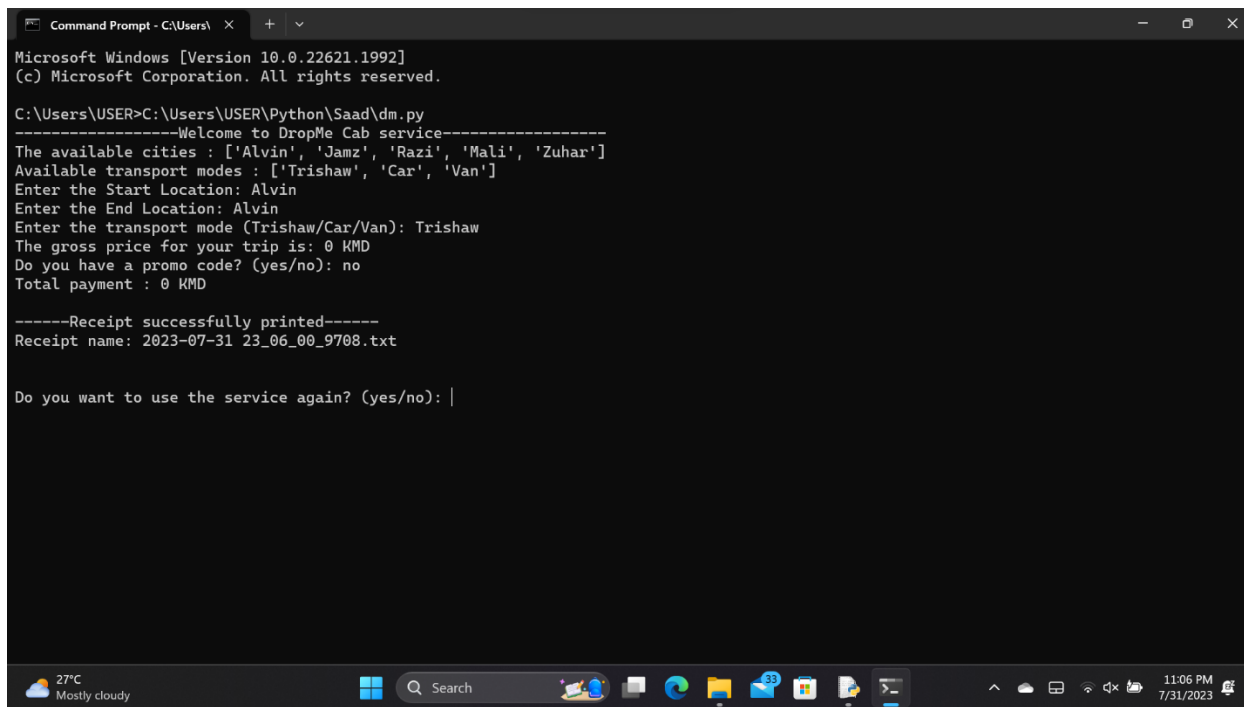
Run the `transport_service()` function within a `if __name__ == "__main__":` block to start the service when the script is executed directly.

4. Test Cases

Test Case Number	Input	Expected Output	Remarks
1	Alvin Alvin	The price between Alvin & Alvin is : 0 KMD.	Fare Calculation between the 2 cities. And creates text file.
2	Alvin Jamz	The price between Alvin & Jamz is : 20 KMD.	Fare Calculation between the 2 cities. And creates text file.
3	Alvin Razi	The price between Alvin & Razi is : 40 KMD.	Fare Calculation between the 2 cities. And creates text file.
4	Alvin Mali	The price between Alvin & Mali is : 40 KMD.	Fare Calculation between the 2 cities. And creates text file.
5	Alvin Zuhar	The price between Alvin & Zuhar is : 20 KMD.	Fare Calculation between the 2 cities. And creates text file.
6	Jamz Alvin / Car	The price between Jamz & Alvin by car is : 40 KMD (2 x 20 KMD)	Fare Calculation between 2 cities by Car. Multiplies the default value by 2. Creates a text file.
7	Jamz Jamz / Car	The price between Jamz & Jamz by car is : 0 KMD (0 x 20 KMD)	Fare Calculation between 2 cities by Car. Multiplies the default value by 2. Creates a text file.
8	Jamz Razi / Car	The price between Jamz & Razi by car is : 40 KMD (2 x 20 KMD)	Fare Calculation between 2 cities by Car. Multiplies the default value by 2. Creates a text file.
9	Jamz Mali / Car	The price between Jamz & Mali by car is : 80 KMD (2 x 40 KMD)	Fare Calculation between 2 cities by Car. Multiplies the default value by 2. Creates a text file.
10	Jamz Zuhar / Car	The price between Jamz & Zuhar by car is : 80 KMD (2 x 40 KMD)	Fare Calculation between 2 cities by Car. Multiplies the default value by 2. Creates a text file.

Table 2 - Test Cases

5. Test Cases Screenshots



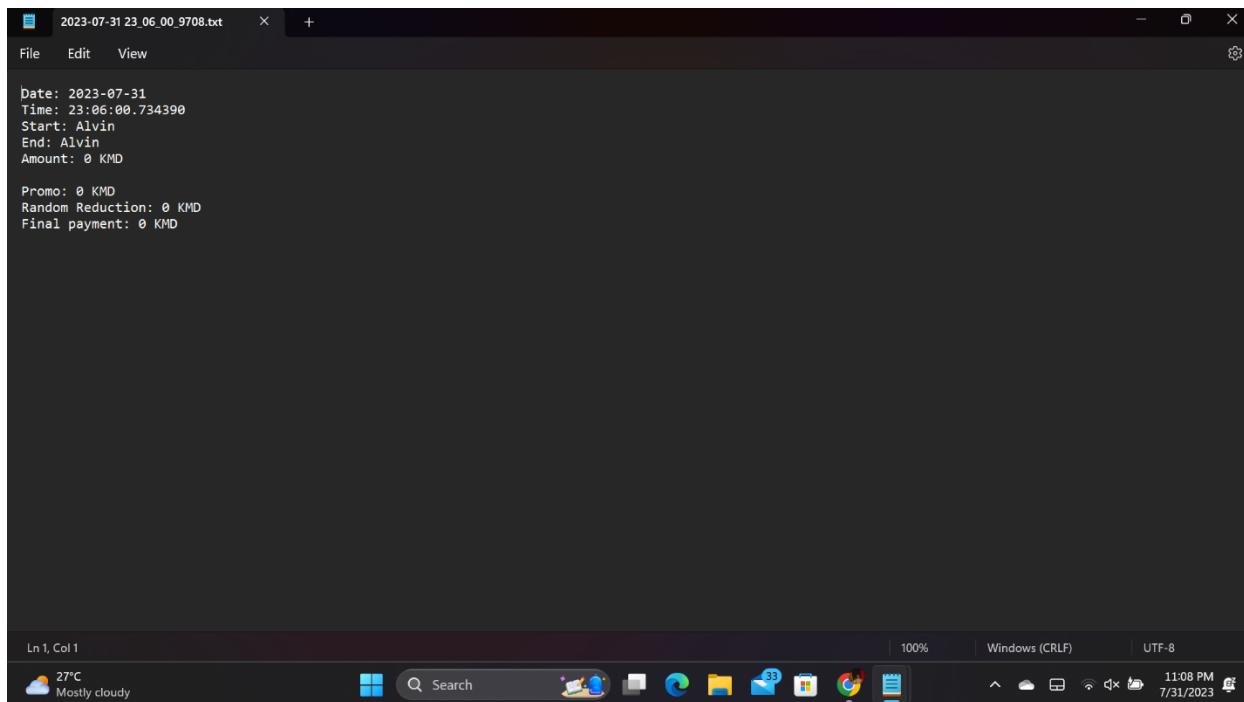
```
Microsoft Windows [Version 10.0.22621.1992]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER>C:\Users\USER\Python\Saad\dm.py
-----Welcome to DropMe Cab service-----
The available cities : ['Alvin', 'Jamz', 'Razi', 'Mali', 'Zuhar']
Available transport modes : ['Trishaw', 'Car', 'Van']
Enter the Start Location: Alvin
Enter the End Location: Alvin
Enter the transport mode (Trishaw/Car/Van): Trishaw
The gross price for your trip is: 0 KMD
Do you have a promo code? (yes/no): no
Total payment : 0 KMD

-----Receipt successfully printed-----
Receipt name: 2023-07-31 23_06_00_9708.txt

Do you want to use the service again? (yes/no): |
```

Figure 1 - Test Case 1



```
2023-07-31 23_06_00_9708.txt
File Edit View

Date: 2023-07-31
Time: 23:06:00.734390
Start: Alvin
End: Alvin
Amount: 0 KMD

Promo: 0 KMD
Random Reduction: 0 KMD
Final payment: 0 KMD

Ln 1, Col 1 100% Windows (CRLF) UTF-8
```

Figure 2 - Test case 1.2


```
Command Prompt - C:\Users\ X + v
The available cities : ['Alvin', 'Jamz', 'Razi', 'Mali', 'Zuhar']
Available transport modes : ['Trishaw', 'Car', 'Van']
Enter the Start Location: Alvin
Enter the End Location: Jamz
Enter the transport mode (Trishaw/Car/Van): Trishaw
The gross price for your trip is: 20 KMD
Do you have a promo code? (yes/no): yes
Enter promo code here: pro5
Total payment : 15 KMD

-----Receipt successfully printed-----
Receipt name: 2023-07-31 21_25_17_3890.txt

Do you want to use the service again? (yes/no): yes
-----Welcome to DropMe Cab service-----
The available cities : ['Alvin', 'Jamz', 'Razi', 'Mali', 'Zuhar']
Available transport modes : ['Trishaw', 'Car', 'Van']
Enter the Start Location: Alvin
Enter the End Location: Jamz
Enter the transport mode (Trishaw/Car/Van): Trishaw
The gross price for your trip is: 20 KMD
Do you have a promo code? (yes/no): no
Total payment : 20 KMD

-----Receipt successfully printed-----
Receipt name: 2023-07-31 23_12_22_3123.txt

Do you want to use the service again? (yes/no): |
```

Figure 3 - Test Case 2.1

```
2023-07-31 23_12_22_3123.txt X +
File Edit View
Date: 2023-07-31
Time: 23:12:22.930708
Start: Alvin
End: Jamz
Amount: 20 KMD

Promo: 0 KMD
Random Reduction: 0 KMD
Final payment: 20 KMD

Ln 1, Col 1 100% Windows (CRLF) UTF-8
```

Figure 4 - Test Case 2.2

```
Command Prompt - C:\Users\ X + v

Do you want to use the service again? (yes/no): yes
-----Welcome to DropMe Cab service-----
The available cities : ['Alvin', 'Jamz', 'Razi', 'Mali', 'Zuhar']
Available transport modes : ['Trishaw', 'Car', 'Van']
Enter the Start Location: Alvin
Enter the End Location: Jamz
Enter the transport mode (Trishaw/Car/Van): Trishaw
The gross price for your trip is: 20 KMD
Do you have a promo code? (yes/no): no
Total payment : 20 KMD

-----Receipt successfully printed-----
Receipt name: 2023-07-31 23_12_22_3123.txt

Do you want to use the service again? (yes/no): yes
-----Welcome to DropMe Cab service-----
The available cities : ['Alvin', 'Jamz', 'Razi', 'Mali', 'Zuhar']
Available transport modes : ['Trishaw', 'Car', 'Van']
Enter the Start Location: Alvin
Enter the End Location: Razi
Enter the transport mode (Trishaw/Car/Van): Trishaw
The gross price for your trip is: 40 KMD
Do you have a promo code? (yes/no): no
Total payment : 40 KMD

-----Receipt successfully printed-----
Receipt name: 2023-07-31 23_14_17_5648.txt

Do you want to use the service again? (yes/no): |
```

Figure 5 - Test Case 3.1

```
2023-07-31 23_14_17_5648.txt X +
File Edit View

Date: 2023-07-31
Time: 23:14:17.018567
Start: Alvin
End: Razi
Amount: 40 KMD

Promo: 0 KMD
Random Reduction: 0 KMD
Final payment: 40 KMD

Ln 1, Col 1 100% Windows (CRLF) UTF-8
27°C Partly cloudy Search 11:14 PM 7/31/2023
```

Figure 6 - Test Case 3.2

```
Command Prompt - C:\Users\ X + v

Do you want to use the service again? (yes/no): yes
-----Welcome to DropMe Cab service-----
The available cities : ['Alvin', 'Jamz', 'Razi', 'Mali', 'Zuhar']
Available transport modes : ['Trishaw', 'Car', 'Van']
Enter the Start Location: Alvin
Enter the End Location: Razi
Enter the transport mode (Trishaw/Car/Van): Trishaw
The gross price for your trip is: 40 KMD
Do you have a promo code? (yes/no): no
Total payment : 40 KMD

-----Receipt successfully printed-----
Receipt name: 2023-07-31 23_14_17_5648.txt

Do you want to use the service again? (yes/no): yes
-----Welcome to DropMe Cab service-----
The available cities : ['Alvin', 'Jamz', 'Razi', 'Mali', 'Zuhar']
Available transport modes : ['Trishaw', 'Car', 'Van']
Enter the Start Location: Alvin
Enter the End Location: Mali
Enter the transport mode (Trishaw/Car/Van): Trishaw
The gross price for your trip is: 40 KMD
Do you have a promo code? (yes/no): no
Total payment : 40 KMD

-----Receipt successfully printed-----
Receipt name: 2023-07-31 23_24_31_7779.txt

Do you want to use the service again? (yes/no): |
```

Figure 7 - Test Case 4.1

```
2023-07-31 23_24_31_7779.txt X +
File Edit View

Date: 2023-07-31
Time: 23:24:31.337432
Start: Alvin
End: Mali
Amount: 40 KMD

Promo: 0 KMD
Random Reduction: 0 KMD
Final payment: 40 KMD

Ln 1, Col 1 100% Windows (CRLF) UTF-8

27°C Partly cloudy Search 11:24 PM 7/31/2023
```

Figure 8 - Test Case 4.2

```
Command Prompt - C:\Users\ X + v

Do you want to use the service again? (yes/no): yes
-----Welcome to DropMe Cab service-----
The available cities : ['Alvin', 'Jamz', 'Razi', 'Mali', 'Zuhar']
Available transport modes : ['Trishaw', 'Car', 'Van']
Enter the Start Location: Alvin
Enter the End Location: Mali
Enter the transport mode (Trishaw/Car/Van): Trishaw
The gross price for your trip is: 40 KMD
Do you have a promo code? (yes/no): no
Total payment : 40 KMD

-----Receipt successfully printed-----
Receipt name: 2023-07-31 23_24_31_7779.txt

Do you want to use the service again? (yes/no): yes
-----Welcome to DropMe Cab service-----
The available cities : ['Alvin', 'Jamz', 'Razi', 'Mali', 'Zuhar']
Available transport modes : ['Trishaw', 'Car', 'Van']
Enter the Start Location: Alvin
Enter the End Location: Zuhar
Enter the transport mode (Trishaw/Car/Van): Trishaw
The gross price for your trip is: 20 KMD
Do you have a promo code? (yes/no): no
Total payment : 20 KMD

-----Receipt successfully printed-----
Receipt name: 2023-07-31 23_26_57_7385.txt

Do you want to use the service again? (yes/no): |
```

Figure 9 - Test case 5.1

```
2023-07-31 23_26_57_7385.txt X +
File Edit View

Date: 2023-07-31
Time: 23:26:57.986127
Start: Alvin
End: Zuhar
Amount: 20 KMD

Promo: 0 KMD
Random Reduction: 0 KMD
Final payment: 20 KMD

Ln 1, Col 1 100% Windows (CRLF) UTF-8

27°C Partly cloudy Search 11:27 PM 7/31/2023
```

Figure 10 - Test Case 5.2

```
Command Prompt - C:\Users\ X + v

Do you want to use the service again? (yes/no): yes
-----Welcome to DropMe Cab service-----
The available cities : ['Alvin', 'Jamz', 'Razi', 'Mali', 'Zuhar']
Available transport modes : ['Trishaw', 'Car', 'Van']
Enter the Start Location: Alvin
Enter the End Location: Zuhar
Enter the transport mode (Trishaw/Car/Van): Trishaw
The gross price for your trip is: 20 KMD
Do you have a promo code? (yes/no): no
Total payment : 20 KMD

-----Receipt successfully printed-----
Receipt name: 2023-07-31 23_26_57_7385.txt

Do you want to use the service again? (yes/no): yes
-----Welcome to DropMe Cab service-----
The available cities : ['Alvin', 'Jamz', 'Razi', 'Mali', 'Zuhar']
Available transport modes : ['Trishaw', 'Car', 'Van']
Enter the Start Location: Jamz
Enter the End Location: Alvin
Enter the transport mode (Trishaw/Car/Van): Car
The gross price for your trip is: 40 KMD
Do you have a promo code? (yes/no): no
Total payment : 40 KMD

-----Receipt successfully printed-----
Receipt name: 2023-07-31 23_29_26_6655.txt

Do you want to use the service again? (yes/no): |
```

Figure 11 - Test Case 6.1

```
2023-07-31 23_29_26_6655.txt X +
File Edit View

Date: 2023-07-31
Time: 23:29:26.001691
Start: Jamz
End: Alvin
Amount: 40 KMD

Promo: 0 KMD
Random Reduction: 0 KMD
Final payment: 40 KMD

Ln 1, Col 1 100% Windows (CRLF) UTF-8

27°C Partly cloudy Search 11:29 PM 7/31/2023
```

Figure 12 - Test Case 6.2

```
Command Prompt - C:\Users\USER>
Microsoft Windows [Version 10.0.22621.1992]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER>C:\Users\USER\Python\Saad\dm.py
-----Welcome to DropMe Cab service-----
The available cities : ['Alvin', 'Jamz', 'Razi', 'Mali', 'Zuhar']
Available transport modes : ['Trishaw', 'Car', 'Van']
Enter the Start Location: Jamz
Enter the End Location: Jamz
Enter the transport mode (Trishaw/Car/Van): Car
The gross price for your trip is: 0 KMD
Do you have a promo code? (yes/no): no
Total payment : 0 KMD

-----Receipt successfully printed-----
Receipt name: 2023-07-31 23_32_34_9282.txt

Do you want to use the service again? (yes/no): |
```

Figure 13 - Test case 7.1

```
2023-07-31 23_32_34_9282.txt
File Edit View

Date: 2023-07-31
Time: 23:32:34.299498
Start: Jamz
End: Jamz
Amount: 0 KMD

Promo: 0 KMD
Random Reduction: 0 KMD
Final payment: 0 KMD

Ln 1, Col 1 100% Windows (CRLF) UTF-8
```

Figure 14 - Test Case 7.2

```
Command Prompt - C:\Users\ \
C:\Users\USER>C:\Users\USER\Python\Saad\dm.py
-----Welcome to DropMe Cab service-----
The available cities : ['Alvin', 'Jamz', 'Razi', 'Mali', 'Zuhar']
Available transport modes : ['Trishaw', 'Car', 'Van']
Enter the Start Location: Jamz
Enter the End Location: Jamz
Enter the transport mode (Trishaw/Car/Van): Car
The gross price for your trip is: 0 KMD
Do you have a promo code? (yes/no): no
Total payment : 0 KMD

-----Receipt successfully printed-----
Receipt name: 2023-07-31 23_32_34_9282.txt

Do you want to use the service again? (yes/no): yes
-----Welcome to DropMe Cab service-----
The available cities : ['Alvin', 'Jamz', 'Razi', 'Mali', 'Zuhar']
Available transport modes : ['Trishaw', 'Car', 'Van']
Enter the Start Location: Jamz
Enter the End Location: Razi
Enter the transport mode (Trishaw/Car/Van): Car
The gross price for your trip is: 40 KMD
Do you have a promo code? (yes/no): no
Total payment : 35 KMD

-----Receipt successfully printed-----
Receipt name: 2023-07-31 23_34_23_7103.txt

Do you want to use the service again? (yes/no): |
```

Figure 15 - Test case 8.1

```
2023-07-31 23_34_23_7103.txt
File Edit View

Date: 2023-07-31
Time: 23:34:23.239613
Start: Jamz
End: Razi
Amount: 40 KMD

Promo: 0 KMD
Random Reduction: 5 KMD
Final payment: 35 KMD

Ln 1, Col 1 100% Windows (CRLF) UTF-8
```

Figure 16 - Test Case 8.2

```
Command Prompt - C:\Users\ >
Do you want to use the service again? (yes/no): yes
-----Welcome to DropMe Cab service-----
The available cities : ['Alvin', 'Jamz', 'Razi', 'Mali', 'Zuhar']
Available transport modes : ['Trishaw', 'Car', 'Van']
Enter the Start Location: Jamz
Enter the End Location: Razi
Enter the transport mode (Trishaw/Car/Van): Car
The gross price for your trip is: 40 KMD
Do you have a promo code? (yes/no): no
Total payment : 35 KMD

-----Receipt successfully printed-----
Receipt name: 2023-07-31 23_34_23_7103.txt

Do you want to use the service again? (yes/no): yes
-----Welcome to DropMe Cab service-----
The available cities : ['Alvin', 'Jamz', 'Razi', 'Mali', 'Zuhar']
Available transport modes : ['Trishaw', 'Car', 'Van']
Enter the Start Location: Jamz
Enter the End Location: Mali
Enter the transport mode (Trishaw/Car/Van): Car
The gross price for your trip is: 80 KMD
Do you have a promo code? (yes/no): no
Total payment : 80 KMD

-----Receipt successfully printed-----
Receipt name: 2023-07-31 23_34_57_7457.txt

Do you want to use the service again? (yes/no): |
```

Figure 18 - Test Case 9.1

```
2023-07-31 23_34_57_7457.txt
File Edit View

Date: 2023-07-31
Time: 23:34:57.624716
Start: Jamz
End: Mali
Amount: 80 KMD

Promo: 0 KMD
Random Reduction: 0 KMD
Final payment: 80 KMD

Ln 1, Col 1 100% Windows (CRLF) UTF-8
```

Figure 17 - Test Case - 9.2